1. Show $\langle \epsilon_a (\epsilon_o - \epsilon_b) \rangle = 0$

analysis error — innovation

★ ALL CODE FOR LATER QUESTIONS HAS BEEN ZIPPED & ATTACHED TO THE SUBMISSION

$$\langle \epsilon_a (\epsilon_o - \epsilon_b) \rangle = \langle (\epsilon_b + w(\epsilon_o - \epsilon_b))(\epsilon_o - \epsilon_b) \rangle$$

$$= \langle \epsilon_b \epsilon_o \rangle^{0} - \langle \epsilon_b^2 \rangle + \langle w(\epsilon_o - \epsilon_b)^2 \rangle$$

$$= -\langle \epsilon_b^2 \rangle + w[\langle \epsilon_o^2 \rangle - 2\langle \epsilon_o \epsilon_b \rangle^{0} + \langle \epsilon_b \rangle^2]$$

$$= -\sigma_b^2 + w[\sigma_o^2 + \sigma_b^2]$$

in lecture, it was shown that the smallest variance occurs for $w = \frac{\sigma_b^2}{\sigma_b^2 + \sigma_o^2}$

in this scalar case, so to minimize variance...

$$\langle \epsilon_a (\epsilon_o - \epsilon_b) \rangle = -\sigma_b^2 + \frac{\sigma_b^2}{\sigma_b^2 + \sigma_o^2}[\sigma_b^2 + \sigma_o^2] = \sigma_b^2 - \sigma_b^2 = 0$$

$$\langle \epsilon_a (\epsilon_o - \epsilon_b) \rangle = 0$$

2. consider a case where $\langle \epsilon_b \rangle = b_b$, $\langle \epsilon_o \rangle = b_o$

then construct unbiased variables $\tilde{x}_o = x_o - b_o$, $\tilde{x}_b = x_b - b_b$

we want our analysis to be unbiased, so write x_a in terms of unbiased vars.

$$\tilde{\epsilon}_b = \tilde{x}_b - x_t = x_b - b_b - x_t = \epsilon_b - b_b \qquad \langle \tilde{\epsilon}_b \rangle = \langle \epsilon_b \rangle - b_b = b_b - b_b = 0$$

$$\tilde{\epsilon}_o = \tilde{x}_o - x_t = x_o - b_o - x_t = \epsilon_o - b_o \qquad \langle \tilde{\epsilon}_o \rangle = \langle \epsilon_o \rangle - b_o = b_o - b_o = 0$$

a) $x_a = \tilde{x}_b + w(\tilde{x}_o - \tilde{x}_b)$

$$x_a - x_t = \tilde{x}_b - x_t + w\left(\tilde{x}_o - x_t - (\tilde{x}_b - x_t)\right)$$

$$\epsilon_a = \tilde{\epsilon}_b + w(\tilde{\epsilon}_o - \tilde{\epsilon}_b)$$

$$\langle x_a - x_t \rangle = \langle \epsilon_a \rangle = \langle \tilde{\epsilon}_b + w(\tilde{\epsilon}_o - \tilde{\epsilon}_b) \rangle = \langle \tilde{\epsilon}_b \rangle + w\langle \tilde{\epsilon}_o \rangle - w\langle \tilde{\epsilon}_b \rangle$$

$$= 0 + 0 + 0 = 0 = \langle \epsilon_a \rangle$$

b) $\langle (\epsilon_a^2) \rangle = \langle \left(\tilde{\epsilon}_b + w(\tilde{\epsilon}_o - \tilde{\epsilon}_b)\right)^2 \rangle$

$$= \langle \tilde{\epsilon}_b^2 + 2w\tilde{\epsilon}_b(\tilde{\epsilon}_o - \tilde{\epsilon}_b) + w^2(\tilde{\epsilon}_o - \tilde{\epsilon}_b)^2 \rangle$$

$$= \langle \tilde{\epsilon}_b^2 \rangle + 2w\langle \tilde{\epsilon}_b \tilde{\epsilon}_o \rangle^{0} - 2w\langle \tilde{\epsilon}_b^2 \rangle + w^2\langle \tilde{\epsilon}_o^2 \rangle - w^2\langle \tilde{\epsilon}_b^2 \rangle$$

(assuming obs. and background are uncorrelated)

$$\langle \epsilon_a^2 \rangle = \langle \tilde{\epsilon}_b^2 \rangle - 2w\langle \tilde{\epsilon}_b^2 \rangle + w^2\left(\langle \tilde{\epsilon}_o^2 \rangle - \langle \tilde{\epsilon}_b^2 \rangle\right)$$

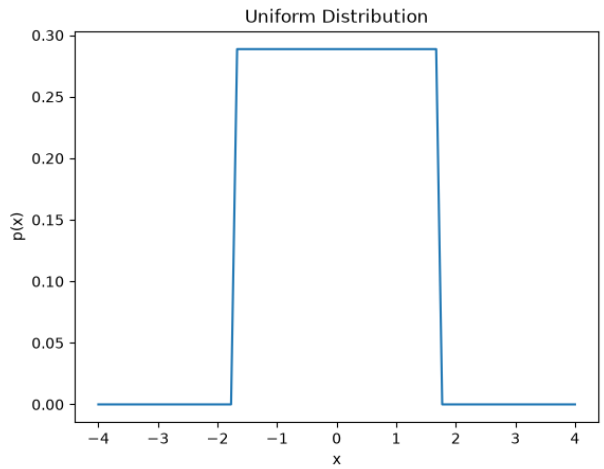
we want... $\dfrac{d\langle \epsilon_a^2 \rangle}{dw} = 0 = 2w\left(\langle \tilde{\epsilon}_o^2 \rangle - \langle \tilde{\epsilon}_b^2 \rangle\right) - 2\langle \tilde{\epsilon}_b^2 \rangle$

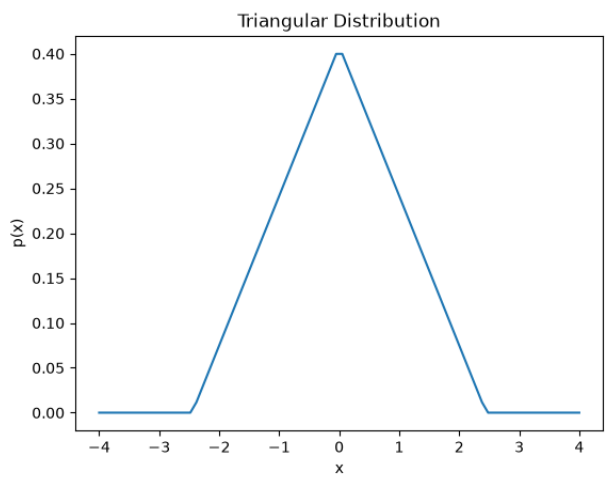
$$\Rightarrow \langle \tilde{\epsilon}_b^2 \rangle = w\left(\langle \tilde{\epsilon}_o^2 \rangle - \langle \tilde{\epsilon}_b^2 \rangle\right)$$

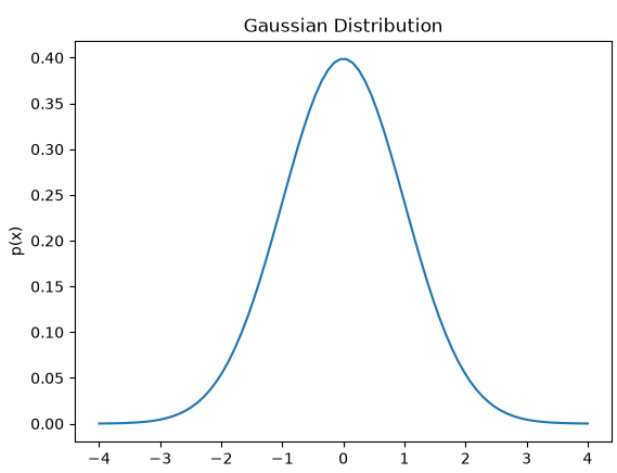
$$w = \langle \tilde{\epsilon}_b^2 \rangle / \left(\langle \tilde{\epsilon}_o^2 \rangle - \langle \tilde{\epsilon}_b^2 \rangle\right)$$

$$w = \frac{\sigma_b^2}{\sigma_o^2 - \sigma_b^2}$$

3. a)


Uniform Distribution
p(x)
x
0.30
0.25
0.20
0.15
0.10
0.05
0.00
−4
−3
−2
−1
0
1
2
3
4


Triangular Distribution
p(x)
x
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
−4
−3
−2
−1
0
1
2
3
4


Gaussian Distribution
p(x)
x
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
−4
−3
−2
−1
0
1
2
3
4

b) uniform

$$\mathcal{E}(x) = \int_{-\infty}^{\infty} x\,p(x)\,dx = \int_{-\sqrt{3}}^{\sqrt{3}} \frac{1}{2\sqrt{3}}\,x\,dx = \frac{1}{2\sqrt{3}}\left[\frac{x^2}{2}\right]_{-\sqrt{3}}^{\sqrt{3}} = 0 = \mathcal{E}(x)$$

$$\mathrm{Var}(x) = \int_{-\infty}^{\infty} x^2 p(x)\,dx = \int_{-\sqrt{3}}^{\sqrt{3}} \frac{1}{2\sqrt{3}}\,x^2\,dx = \frac{1}{2\sqrt{3}}\left[\frac{x^3}{3}\right]_{-\sqrt{3}}^{\sqrt{3}}$$

$$= \frac{1}{2(3)^{3/2}}\left[3^{3/2} + 3^{3/2}\right] = 1$$

$\mathrm{Var}(x) = 1$

triangular:

we note that this function is symmetrical about $x=0$. this means that $\mathcal{E}(x) = 0$

$$\mathrm{Var}(x) = \int_{-\infty}^{\infty} x^2\,p(x)\,dx = \int_0^{\sqrt{6}} x^2\,\frac{\sqrt{6}-x}{6}\,dx - \int_0^{-\sqrt{6}} x^2\,\frac{\sqrt{6}+x}{6}\,dx$$

$$= \frac{1}{6}\left(\int_0^{\sqrt{6}} (\sqrt{6}\,x^2 - x^3)\,dx - \int_0^{-\sqrt{6}} (\sqrt{6}\,x^2 + x^3)\,dx\right)$$

$$= \frac{1}{6}\left(\left[\sqrt{6}\,\frac{x^3}{3} - \frac{x^4}{4}\right]_0^{\sqrt{6}} - \left[\sqrt{6}\,\frac{x^3}{3} + \frac{x^4}{4}\right]_0^{-\sqrt{6}}\right) = \frac{1}{6}\left(12 - 9 - (-12) - 9\right) = \frac{6}{6} = 1$$

$\mathrm{Var}(x) = 1$

Gaussian

we note that this function is symmetrical about $x=0$. this means that $\mathcal{E}(x) = 0$

$\mathrm{Var}(x) = \int_{-\infty}^{\infty} x^2 p(x)\,dx = \int_{-\infty}^{\infty} x^2 \left(\frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right)\right) dx$, since this is a symmetric function, $\mathrm{Var}(x) = \frac{2}{\sqrt{2\pi}} \int_0^{\infty} x^2 \exp\left(-\frac{x^2}{2}\right) dx = \sqrt{\frac{2}{\pi}} \int_0^{\infty} x^2 \exp\left(-\frac{x^2}{2}\right) dx$

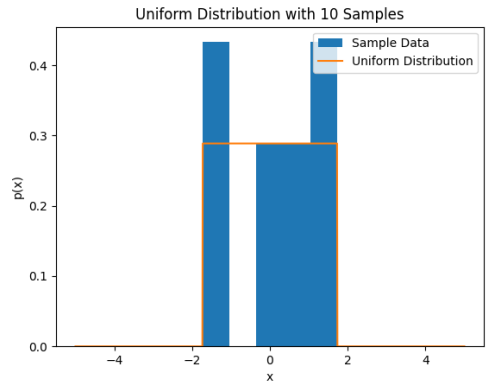
choose $u = \frac{x^2}{2}$ $\quad x = \sqrt{2u} \Rightarrow dx = \frac{1}{\sqrt{2u}}\,du$

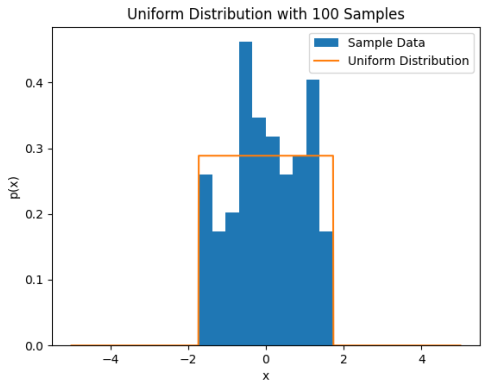
$$\mathrm{Var}(x) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} 2u \exp(-u)\,\frac{1}{\sqrt{2u}}\,du = \sqrt{\frac{2}{\pi}} \int_0^{\infty} \sqrt{2}\sqrt{u} \exp(-u)\,du$$

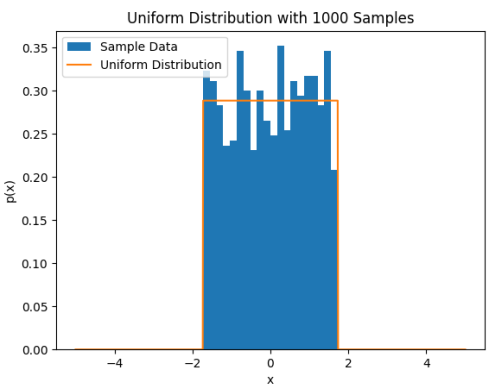
$$= \sqrt{\frac{2}{\pi}}\sqrt{2}\,\Gamma\left(\frac{3}{2}\right)$$ then since $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$ and $\Gamma(s+1) = s\Gamma(s)$

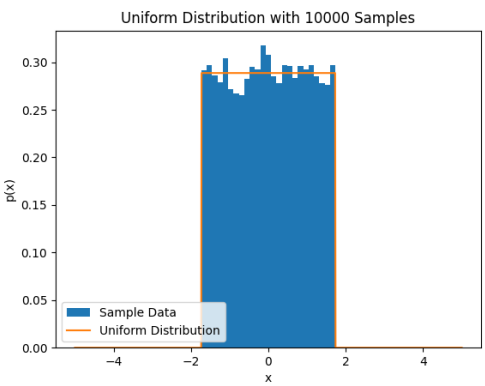
$$= \frac{2}{\sqrt{\pi}}\Gamma\left(\frac{3}{2}\right) = \frac{2}{\sqrt{\pi}}\sqrt{\pi}\cdot\frac{1}{2} = 1$$ $\mathrm{Var}(x) = 1$

c) a "good distribution" is very subjective, but for a uniform distribution, it seems a lot of samples are required, the distribution begins to look more uniform with 1000 samples, but even 10,000 still gives a notable improvement over 1,000 samples. (plots on next page)



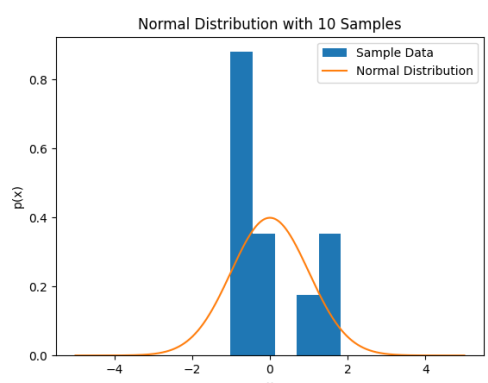


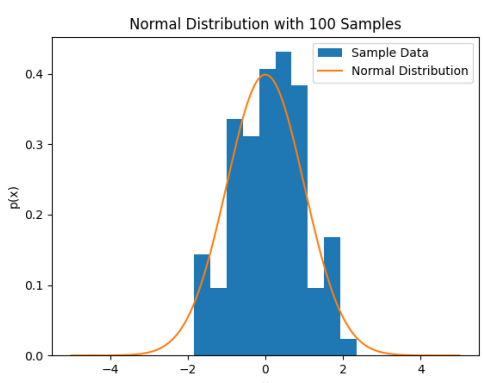


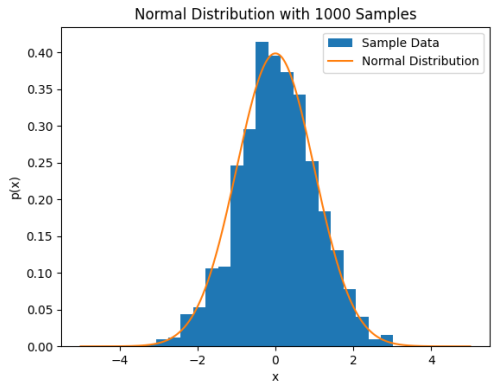


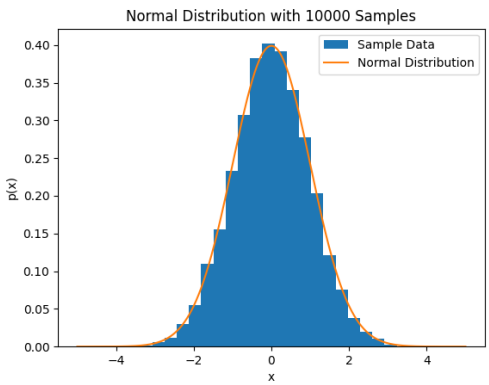
d) In comparison to the uniform distribution, the Gaussian distribution seems to converge for much smaller numbers of samples. by 100 it already has a shape much like the distribution, and by 1000 it is a very good approximation.

(plots on next page)

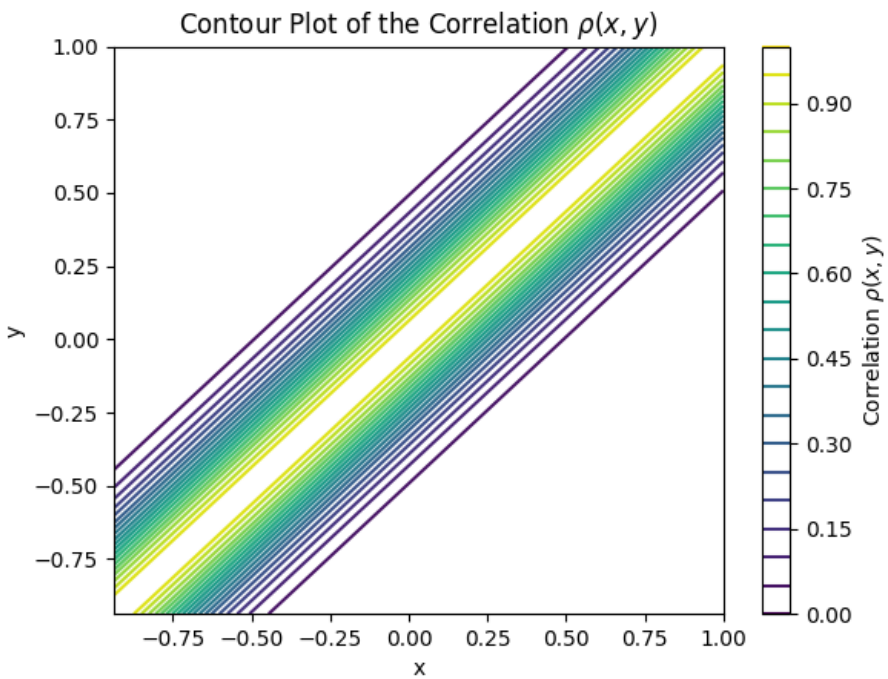




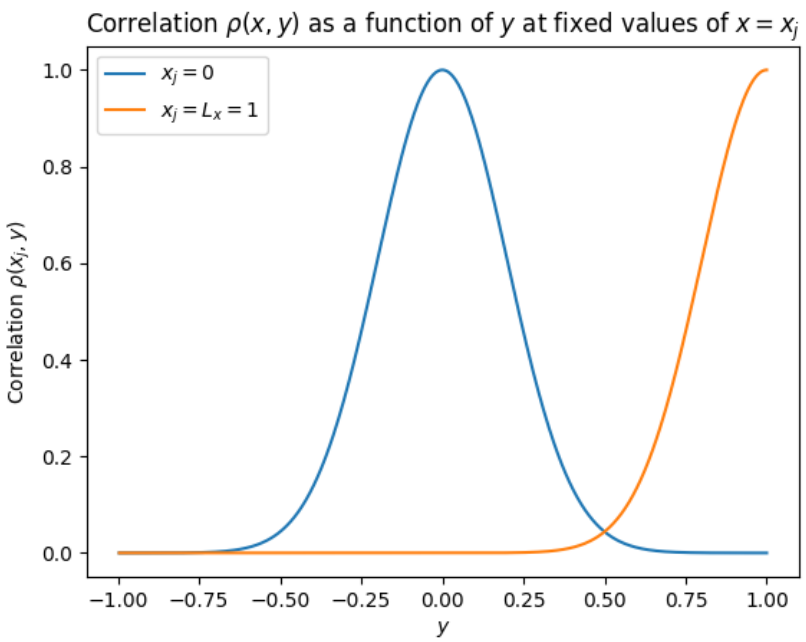

Normal Distribution with 1000 Samples
Sample Data
Normal Distribution
p(x)
x
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0.40
−4
−2
0
2
4


Normal Distribution with 10000 Samples
Sample Data
Normal Distribution
p(x)
x
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0.40
−4
−2
0
2
4

4. a)



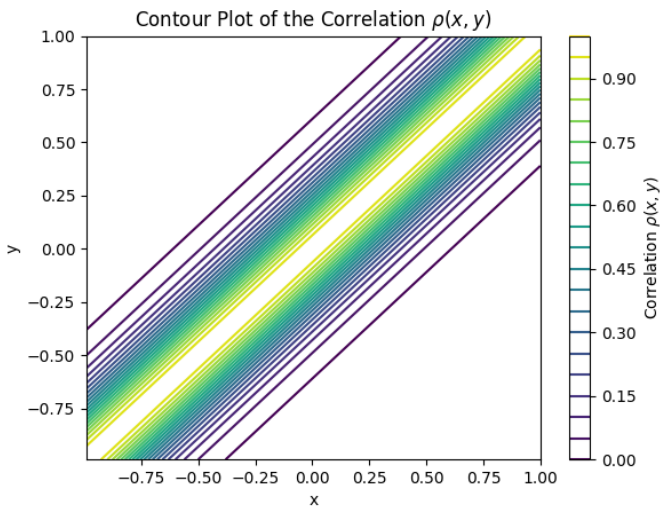
b)



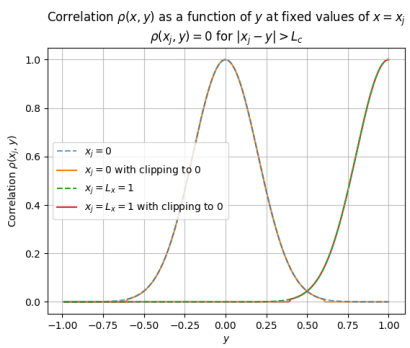
c) "acceptable correlation" depends on the application, but for our purposes in

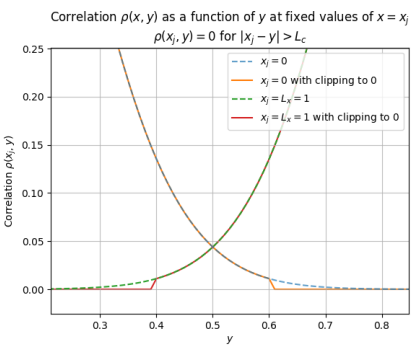
many DA models we want $\max(\rho(x,y)) = 1$, and both $\lim_{x \to \pm\infty} \rho(x,y) = 0$ and $\lim_{y \to \pm\infty} \rho(x,y) = 0$ since measurements with infinite separation will typically have low correlation in these types of systems. with this taken into account, this is an acceptable metric for correlation

d)



there is not a super obvious change from the contour plot since at L_c, the value is already very close to zero.

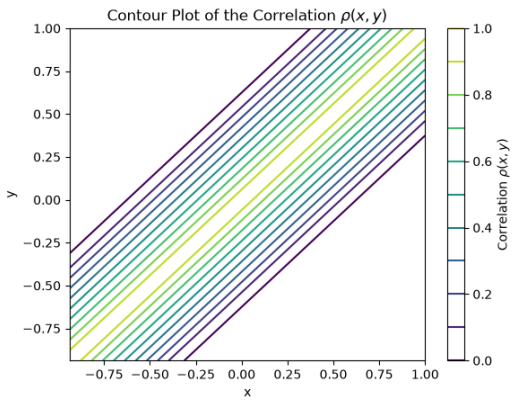




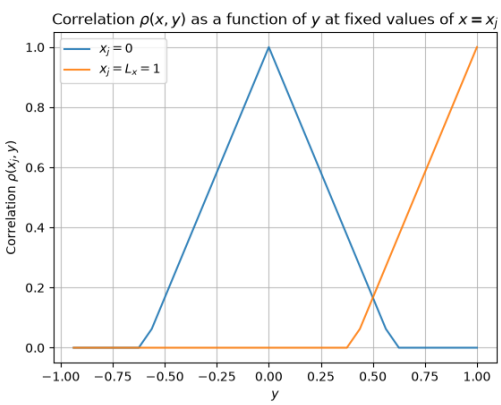
when plotting the correlation functions from the same 2 locations as in part b), we can see where the cutoff occurs (especially when we zoom in)

this is an acceptable correlation metric based on the criteria defined above

e)



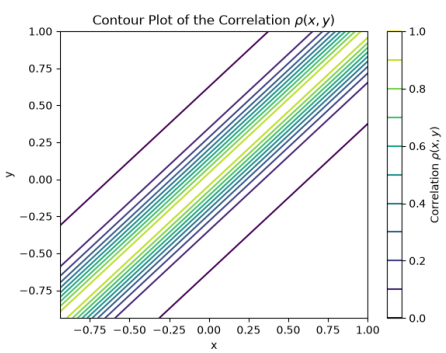
→ plot from part a.



→ plot from part b.

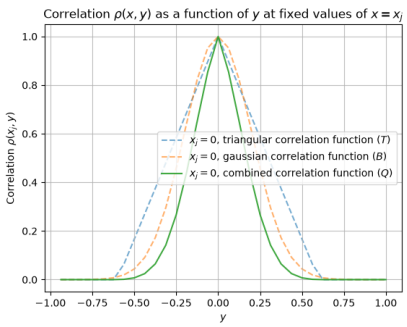
T could be an acceptable correlation matrix because it is finite at infinite distance, and has values of $\rho \in [0,1]$.

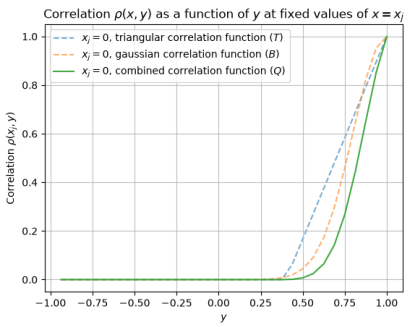
f)



→ contour plot for $Q = B \circ T$

→ we see interestingly that the value does go to zero

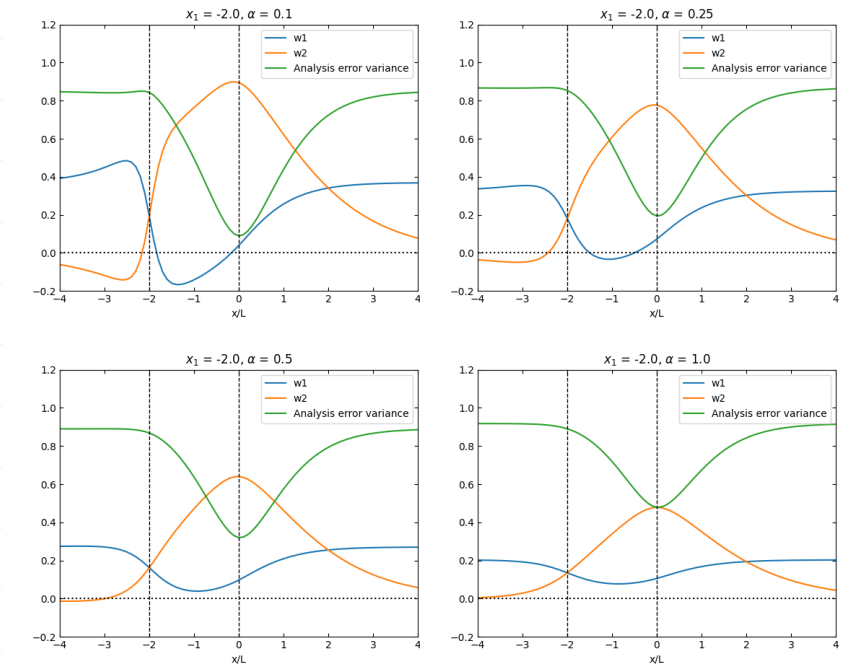




I believe Q is an acceptable correlation matrix for the reasons described above. the differences in the three matrices can be seen above, and the plot on the right shows how Q and B are both finite as measurements get very far away from each other, which is typically what we would want in a DA correlation matrix.

5. i) $\alpha = \frac{\sigma_r^2}{\sigma_b^2}$, as $\alpha \uparrow$, background has higher weight in analysis
as $\alpha \downarrow$, observation has higher weight in analysis

taking the values $\alpha = [0.1, 0.25, 0.5, 1.0]$ gives the following plots



a) as α increases, so does the analysis error.

α	0.1	0.25	0.5	1.0
ϵ_a^2 (minimum value)	0.08937	0.1938	0.3197	0.4781

this is happening because of the form of ϵ_a^2 implemented in the given python script. In the script, there is no information about either σ_r^2 or σ_b^2. As a result, ϵ_a^2 is calculated from the following expression

$$\epsilon_a^2 = \left(\frac{\sigma_a}{\sigma_b}\right)^2 = \left(1 - \frac{(1+\alpha)(\rho_{10}^2 + \rho_{20}^2) - 2\rho_{10}\rho_{20}\rho_{12}}{(1+\alpha)^2 - \rho_{12}^2}\right)$$

from this expression, we can obviously see directly that as α increases, ϵ_a^2 should increase towards $\epsilon_a^2 = 1$.

In terms of the analysis of our system, we can interpret this by thinking about the components of both $\alpha = \left(\frac{\sigma_r^2}{\sigma_b^2}\right)$ and $\epsilon_a^2 = \left(\frac{\sigma_a^2}{\sigma_b^2}\right)$. as α increases, if σ_b^2 decreasing is what drives the increase in α then there will be an equivalent scaling of ϵ_a^2; however, if it is σ_r^2 increasing that drives the increase in α, then it makes sense for our analysis error to increase since the quality of our observations has decreased, and they will be less able to inform our analysis.

b) as α increases, observation screening decreases. This occurs because of the effective increase of σ_r^2 relative to σ_b^2 that is the result of α increasing.

→ For small α, (as in the top left plot) the screening effect is extremely strong because there is a very trustworthy measurement nearer to the analysis than the observation which is being screened.

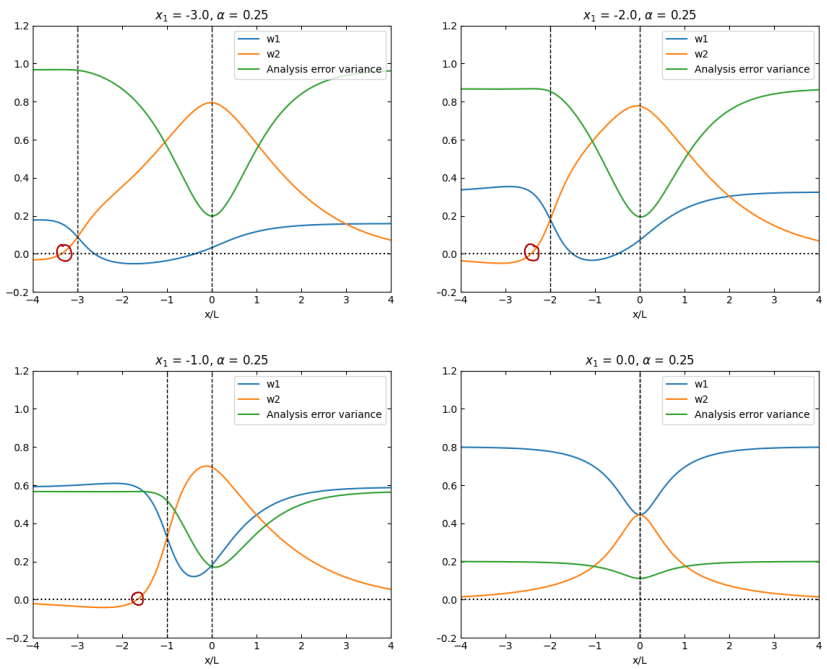
→ whichever observation is slightly further from the analysis is negative

→ there is an equilibrium where both weights are the same which occurs when the observations are at the same point.

→ for large α (as in the bottom right plot), the degree of relative precision of the observations is much lower so the additional information of slightly further away observations

is still useful to constrain our analysis.

ii)



x_1	-3	-2	-1	0
ϵ_a^2 (minimum value)	0.1987	0.1938	0.1699	0.1111

a) when observation 1 is at x_1, observations 1 and 2 have equal weights only when observation 2 is at $x = \pm x_1$

b) The locations where observation 2 has no effect on the analysis are the roots of w_2. These are shown below.

x_1	-3	-2	-1	0
x_2 where $w_2 = 0$	-3.364	-2.429	-1.667	No roots.

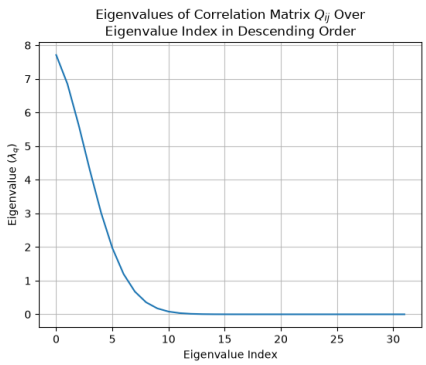
c) the best location for obs. 1 is at x=0. ← This minimizes the analysis error (as shown in the table above). (at the analysis point)

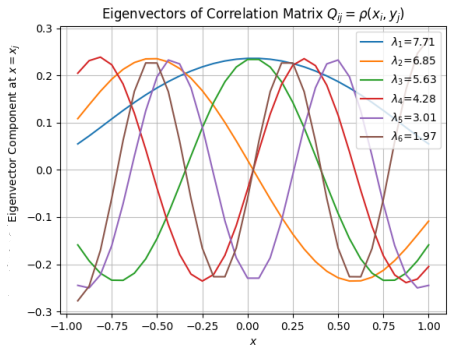
↳ the contribution of observation 2 decays exponentially with distance away from x=0 in this case, and never actually goes to 0.

↳ the analysis error does not appreciably change anymore after r/x=3, so that is the limit of real influence of the second observation on the analysis.

6. $Q_{ij} = \rho(x_i, y_j) = \exp\left(-\frac{1}{2}\frac{(x-y)^2}{L^3}\right)$

a)

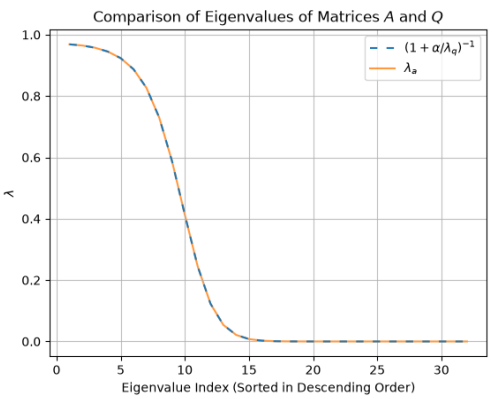




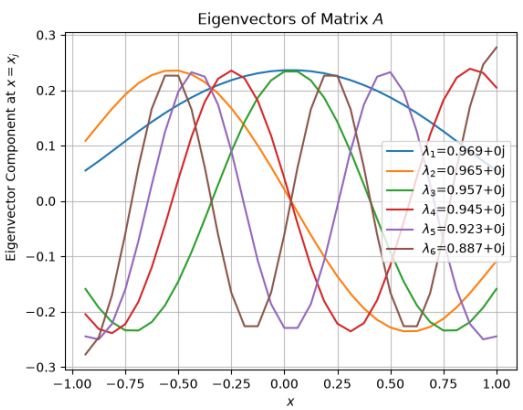
→ these are the eigenvectors associated with the 6 largest eigenvalues.
→ # of crossings in table.

eigenvalue	λ_1	λ_2	λ_3	λ_4	λ_5	λ_6
# of 0 crossings	0	1	2	3	4	5

b)

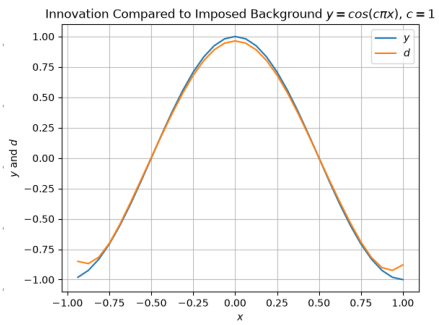


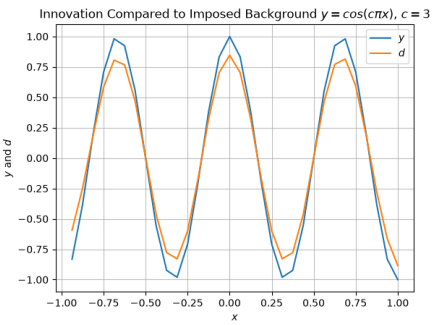
→ the eigenvalues of A are equal to $(1+\alpha/\lambda_q)^{-1}$ where λ_q are the eigenvalues of q

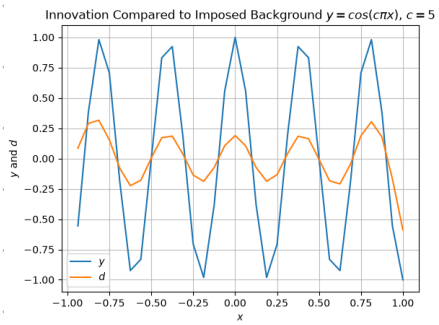


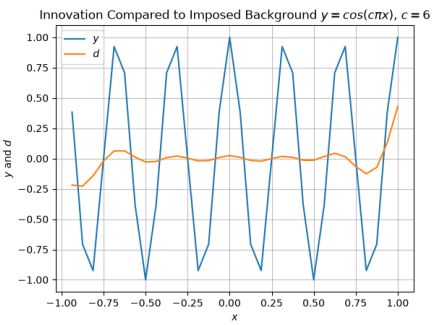
→ the eigenvectors of A form the same basis as the eigenvectors of Q, but have different weights (in the form of their eigenvalues)

c)







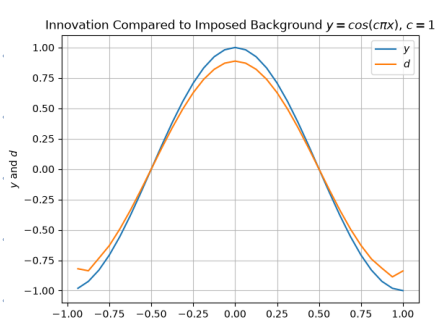


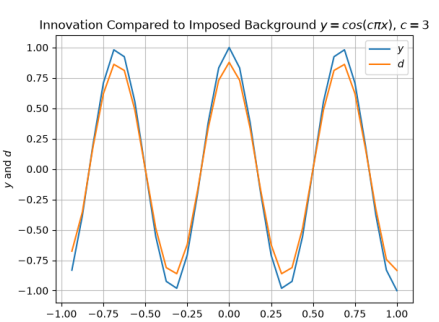
in the plots above, we can see that the waves associated with low wavenumber modes are filtered the least, with higher wavenumber modes being filtered more.

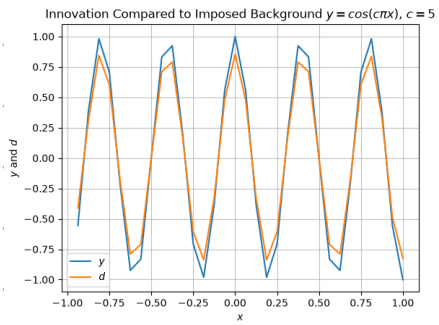
→ this makes sense, as the lengthscales that we can resolve (based on the correlation matrix) are quite large.

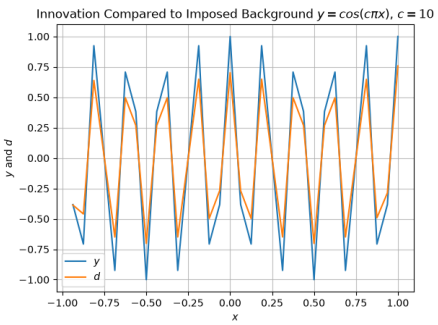
→ all waves with integer wavenumbers $c \geq 6$ have their amplitude reduced by more than 80%.

d)









The plots above have $L_d = 0.05$ rather than $L_d = 0.2$ (which was the value for the plots in 6.c) this means the covariance lengthscale is much smaller, and we should be able to confidently resolve smaller scales. This means that higher wavenumber waves contribute more to our analysis than they did previously.

→ even wavenumbers higher than $C = 10$ still contribute substantially to the solution (See above)

→ the opposite effect happens when L_d is increased.

→ as $L_d \to 0$, $\rho \to 0$ faster, so smaller scale effects are accounted for.